FoodBridge - Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for FoodBridge, a web-based food rescue and donation platform. The system helps restaurants, hotels, event organizers, NGOs, volunteers, and administrators coordinate surplus food donation, request approval, pickup, delivery, tracking, and reporting.

1.2 Scope

FoodBridge supports secure user registration and login, role-based access, food donation posting, NGO food requests, donor approval, volunteer delivery task management, real-time status tracking, notifications, NGO verification, user management, and reporting. The system focuses on reducing food waste and improving food distribution to people in need

Functional Requirements:

FR-01: User Registration

The system shall allow donors, NGO representatives, volunteers, and administrators to register an account using valid information.

FR-02: User Login

The system shall allow registered users to log in using valid email/username and password.

FR-03: Role-Based Dashboard

The system shall show different dashboards based on user roles such as donor, NGO, volunteer, and administrator.

FR-04: Profile Management

The system shall allow users to view and update their profile information.

FR-05: Create Food Donation

The system shall allow donors to create food donation posts by entering food type, quantity, expiration time, and pickup location.

FR-06: Edit or Delete Donation

The system shall allow donors to edit or delete their food donation posts before approval or pickup.

FR-07: View Available Donations

The system shall allow NGOs to browse and view available food donations.

FR-08: Search and Filter Donations

The system shall allow NGOs to search or filter donations by food type, location, quantity, and expiration time.

FR-09: Submit Food Request

The system shall allow NGOs to submit requests for suitable food donations.

FR-10: Approve or Reject Request

The system shall allow donors to approve or reject food requests submitted by NGOs.

FR-11: Volunteer Task View

The system shall allow volunteers to view available delivery tasks.

FR-12: Accept Delivery Task

The system shall allow volunteers to accept a delivery task based on availability.

FR-13: Pickup Confirmation

The system shall allow volunteers to update the delivery status after collecting food from the donor.

FR-14: Delivery Tracking

The system shall track food delivery progress from pickup location to NGO/distribution center.

FR-15: Delivery Confirmation

The system shall allow NGOs to confirm food receipt after successful delivery.

FR-16: Notification Management

The system shall send notifications for donation requests, approvals, volunteer assignments, pickup updates, and delivery completion.

FR-17: Donation Status Update

The system shall update donation status such as Pending, Requested, Approved, Picked Up, Delivered, and Completed.

FR-18: Admin User Management

The system shall allow administrators to manage donors, NGOs, volunteers, and other users.

FR-19: NGO Verification

The system shall allow administrators to verify NGO accounts before they can request food donations.

FR-20: Report Generation

The system shall allow administrators to generate reports about donations, requests, deliveries, and user activity.

Non-Functional Requirements:

NFR-01: Security

The system shall protect user data using secure authentication and authorization.

NFR-02: Role-Based Access Control

The system shall ensure that users can access only the features allowed for their role.

NFR-03: Data Privacy

The system shall keep personal information, donation records, and delivery details private from unauthorized users.

NFR-04: Performance

The system shall load donation lists, user dashboards, and request pages within a reasonable time.

NFR-05: Reliability

The system shall store donation, request, and delivery records reliably without data loss.

NFR-06: Availability

The system shall be available to users most of the time so that donations and deliveries can be managed without interruption.

NFR-07: Usability

The system shall provide a simple and user-friendly interface for donors, NGOs, volunteers, and administrators.

NFR-08: Mobile Responsiveness

The system shall work properly on mobile, tablet, and desktop devices.

NFR-09: Scalability

The system shall support an increasing number of users, donations, requests, and delivery records in the future.

NFR-10: Maintainability

The system code and documentation shall be organized so that developers can update and maintain the system easily.

NFR-11: Accuracy

The system shall show correct donation status, request status, and delivery updates.

NFR-12: Notification Timeliness

The system shall send important notifications to users on time.

NFR-13: Backup and Recovery

The system shall support backup and recovery to prevent permanent data loss.

NFR-14: Transparency

The system shall keep proper records of donation activities to maintain trust among donors, NGOs, volunteers, and administrators.

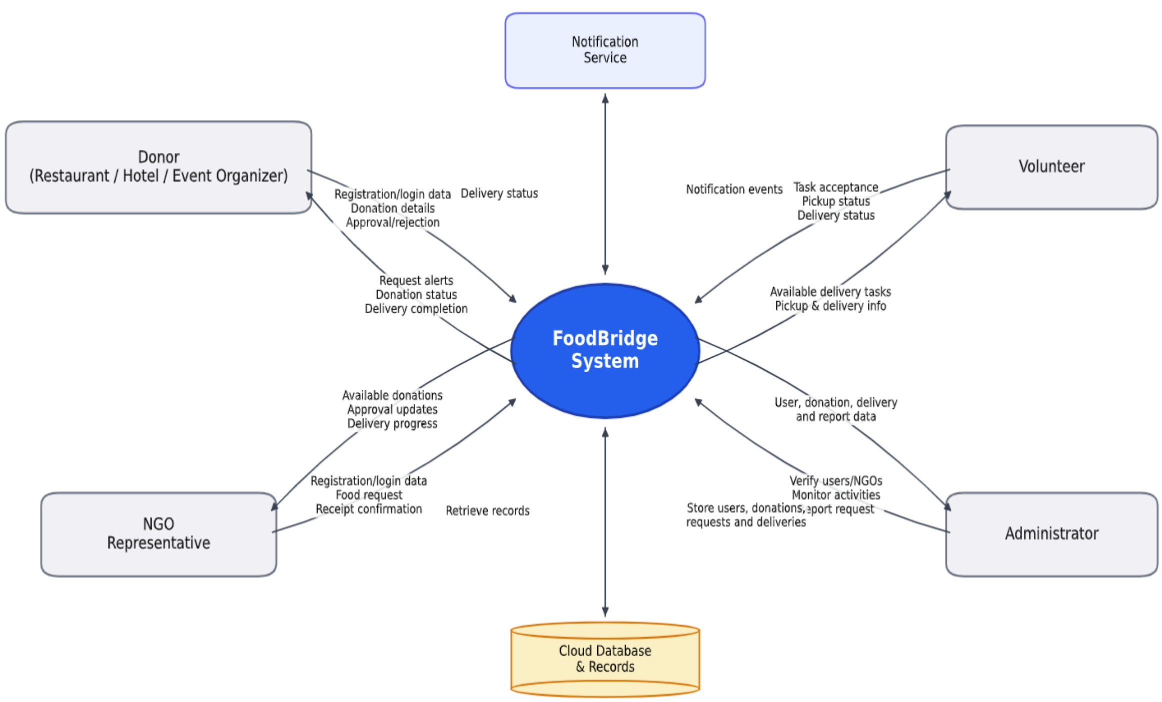
NFR-15: Compatibility

The system shall work on commonly used modern web browsers.

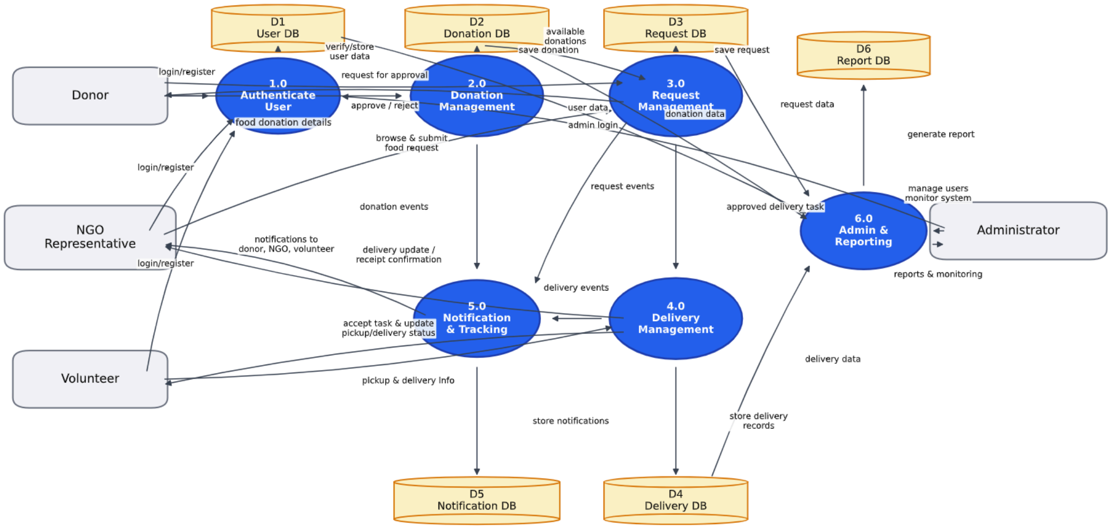
FoodBridge App — Data Flow Diagrams (DFD):

Dataflow Diagram (DFD)
The Data Flow Diagram represents how data moves between the FoodBridge system, users, external services, and databases.

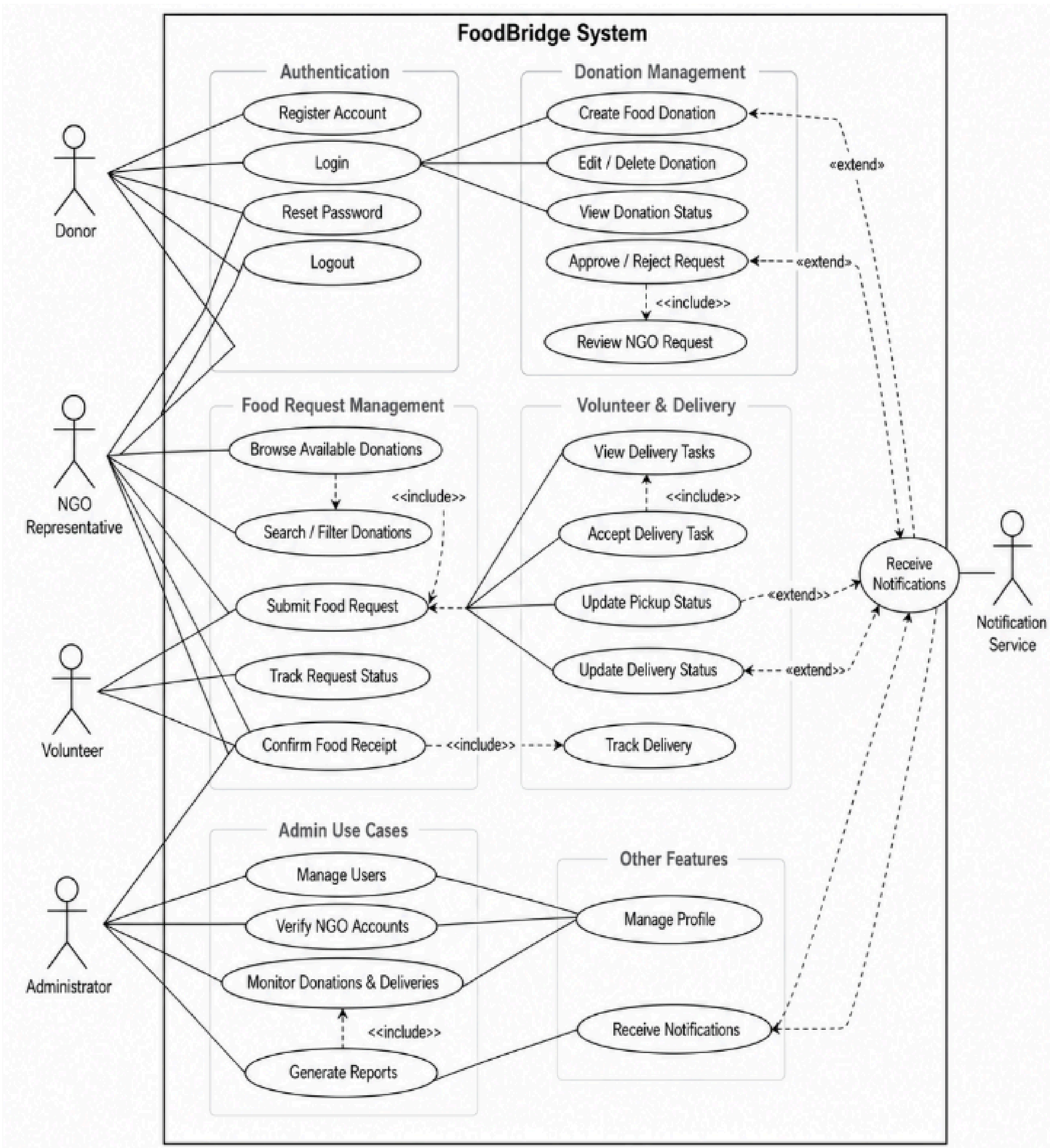
Level 0 DFD Context Diagram
The Level 0 DFD shows FoodBridge as a single main system interacting with donors, NGO representatives, volunteers, administrators, the cloud database, and the notification service. It defines the system boundary and high-level data flows entering and leaving the system.



Level 1 DFD — Main Processes
The Level 1 DFD decomposes FoodBridge into six core processes: Authenticate User, Donation Management, Request Management, Delivery Management, Notification & Tracking, and Admin & Reporting. It also shows the main data stores such as User DB, Donation DB, Request DB, Delivery DB, Notification DB, and Report DB.



FoodBridge —- Use Case Diagram :



External Interface Requirements:

User Interfaces

The system shall provide responsive web interfaces for donors, NGOs, volunteers, and administrators. Each user role shall have a dashboard with relevant actions and information. The interface shall be easy to use and accessible from mobile and desktop devices.

Software Interfaces

- Authentication service for registration and login
- Database service for storing user, donation, request, delivery, notification, and report data
- Notification service for alerts and status updates
- Map/location service may be used for pickup and delivery location support
- REST API or similar backend service for frontend-backend communication

Hardware Interfaces

The system does not require special hardware. Users can access the platform using smartphones, tablets, laptops, or desktop computers with internet access.

Communication Interfaces

- HTTPS for secure communication
- JSON data format for API requests and responses
- Email, SMS, or in-app notification channel for important system updates

Assumptions and Constraints

1. The MVP focuses only on food donation and distribution management.
2. AI food prediction and government integration are outside the current release scope.
3. Verified NGOs are required for safe and trustworthy food distribution.
4. Donors are responsible for providing accurate food quantity, type, expiration time, and pickup location.
5. Volunteers are responsible for updating delivery status after pickup and delivery.
6. The system must maintain data security, reliability, and transparency.
7. The project timeline is limited to academic milestones and releases.